

Push Button Controlled LED using ESP32

Project Overview

This project demonstrates how to interface a push button with an ESP32 DevKit V1 to control an LED. The ESP32 monitors the push button continuously. When pressed, the LED turns ON; when released, the LED turns OFF. The project also introduces the ESP32 internal pull-up resistor (INPUT_PULLUP).

Objective

- Understand digital input and output.
- Interface a push button with ESP32.
- Learn INPUT_PULLUP.
- Read digital signals using digitalRead().
- Control an LED using conditional statements.

Components Used

Component	Quantity
ESP32 DevKit V1	1
Breadboard	1
Push Button	1
LED	1
220 Ω Resistor	1
Jumper Wires	As Required
USB Cable	1

Pin Configuration

LED → GPIO 2

Push Button → GPIO 4

Circuit Description

GPIO 2 is configured as OUTPUT for the LED. GPIO 4 is configured as INPUT_PULLUP. When the button is not pressed, GPIO 4 reads HIGH. Pressing the button connects GPIO 4 to GND, making it LOW.

Working Principle

The ESP32 continuously reads the button state. If the button is LOW, the LED turns ON. If the button is HIGH, the LED turns OFF.

Flow of Execution

1. Configure GPIO 2 as OUTPUT.
2. Configure GPIO 4 as INPUT_PULLUP.
3. Read GPIO 4.
4. If LOW → LED ON.
5. Else → LED OFF.
6. Repeat continuously.

Programming Concepts Used

GPIO Programming, pinMode(), digitalRead(), digitalWrite(), INPUT_PULLUP, Conditional Statements.

Learning Outcomes

GPIO Programming, Digital Input, Digital Output, Internal Pull-Up Resistor, Push Button Interfacing, Basic Embedded Systems.

Applications

User input systems, electronic switches, home automation, embedded systems learning.

Future Improvements

Software debouncing, interrupts, multiple buttons, OLED display, wireless control.

Conclusion

This project demonstrates digital input handling using the ESP32 and provides a solid foundation for more advanced embedded applications involving buttons and switches.

Author

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Board: ESP32 DevKit V1 (30-Pin)
Programming Language: Arduino (C++)
Development Platform: Arduino IDE